Software Test Plan (STP)

AI-Powered Content Management System (AICMS)

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Submission boundary: This submission defines requirements, proposed design, and planned verification. It does not claim that a working implementation exists or that tests have been executed.

Document Control

Version	Date	Author	Status
1.0	09-10-2026	AICMS Team (Shreeverdhan M, Sripada Krishna Sameer Aditya, Srivamshi Chikkireddi, Shamanth SK)	Stage 1 plan; test execution pending

1. Introduction

1.1 Purpose

This Software Test Plan defines planned verification for AICMS, including scope, approach, environment, schedule, roles, risks, test data, case traceability, and exit criteria. It is a planning artifact; no implementation or execution result is claimed.

1.2 Scope

Testing is planned for document upload/validation, duplicate handling, text extraction/OCR, AI category and keyword/topic suggestions, embeddings, indexing, semantic search, filtering/ranking, metadata correction, authorization, audit, administration, deletion consistency, performance, accessibility, and error recovery.

1.3 References and Definitions

References: AICMS SRS v1.0 and AICMS Software Architecture and Design Specification v1.0. Definitions: SRS – Software Requirements Specification; STP – Software Test Plan; RTM – Requirements Traceability Matrix; OCR – Optical Character Recognition; RBAC – Role-Based Access Control; Precision@5 – fraction of the top five retrieved results judged relevant.

2. Test Items

- Web UI: upload, processing status, search, result display, metadata review, administrator views.
- Application APIs: document upload/detail, search, metadata update, re-index, deletion.
- Intake/validation, file/object storage, metadata database, and duplicate policy.

- Text extraction/OCR and asynchronous processing queue/worker.
- AI enrichment: categories, keywords/topics, embeddings, failure states.
- Search/retrieval: lexical/vector index, ranking, filters, authorization-before-disclosure; security/audit, metrics, logs, configuration, and recovery paths.

3. Features to be Tested

Feature / SRS requirement IDs	Planned verification
AICMS-F-001–004	Valid upload, invalid/oversize file, duplicate policy, visible processing states.
AICMS-F-005–007	Text extraction, metadata extraction, OCR behavior and quality flags.
AICMS-F-008–010	Category suggestions, keyword/topic output, embedding creation and model-failure behavior.
AICMS-F-011–015	Indexing, natural-language query, ranking, filters, results/snippets.
AICMS-F-016–018, 023; AICMS-SR-001,004,006	Metadata correction, login/logout, access enforcement, audit completeness, no search disclosure.
AICMS-F-019–022, 024	Taxonomy changes, re-index jobs, deletion/index consistency, operational metrics, user and role administration.
AICMS-NF-001–010; AICMS-SR-002,003,005,007,008	Latency, asynchronous responsiveness, accessibility, data integrity, secure uploads, TLS, and secret/log hygiene; categorization accuracy (≥ 80%), 10-page processing time (≤ 60 s), password storage, session expiry and login lock-out.

4. Features Not to be Tested

- Correctness of third-party model internals beyond system-level evaluation of observable AICMS outputs.
- Provider internals of external identity, storage, model, or search services beyond their integration contracts.
- Production monthly availability before a representative deployment and observation period exist.
- File formats and languages explicitly outside the agreed supported scope.
- Real sensitive production documents; tests use synthetic or formally approved non-sensitive data.

5. Test Approach / Strategy

5.1 Test Levels and Types

- Unit tests: validators, metadata rules, permission decisions, ranking/filter logic, job-state transitions.
- Integration tests: API ↔ storage/database, parser/OCR adapter, model interface, queue/worker, and search index.
- System/end-to-end tests: upload → process → search → metadata edit → re-index/delete.
- Regression tests: re-run the affected smoke and high-priority cases after each defect fix or model/index configuration change.
- Acceptance tests (UAT): project representative runs the end-to-end flows against the acceptance criteria in SRS section 6.

- Negative tests: malformed/unsupported files, empty query, invalid payload, missing role, denied document, and model/index failure.
- Security tests: role bypass, direct document-ID access, restricted snippet leakage, file validation, TLS, secret/log inspection, rate limits.
- Performance tests: p90 search latency, concurrency, queue delay, large permitted upload, and job throughput.
- Accessibility tests: keyboard-only use, visible focus, labels, contrast, and safe errors in core workflows.
- AI/search quality tests: labelled query-document corpus; review category/keyword outcomes and report Precision@5 and relevant-document coverage.

5.2 Entry Criteria

- SRS, SAD, and test cases are reviewed and the target behavior is agreed.
- A testable implementation build and isolated environment are available in a later stage.
- Synthetic files, role-specific test accounts, expected metadata, and labelled search judgments are prepared.
- Required model/search/storage integrations or deterministic service stubs are configured securely.
- File types, size limits, languages, search benchmark, performance load, and accepted quality thresholds are documented.

5.3 Exit Criteria

- 100% of planned high-priority cases are executed or have an approved waiver with rationale.
- No unresolved critical defect remains; high-severity issues are dispositioned by the project representative.
- All high-priority functional and security requirements pass their acceptance criteria.
- No unresolved critical/high access-control or data-disclosure issue remains.
- Performance and search-quality results are recorded against the named dataset/environment; categorization accuracy ≥ 80% and a relevant document in the top 5 results for ≥ 80% of benchmark queries are achieved.
- RTM includes a status/evidence reference for every in-scope requirement.

5.4 Security Validation

- Validate authentication and role/document authorization, including direct ID, result title, snippet, metadata, edit, and deletion attempts.
- Verify unsupported, oversize, malformed, and disguised file handling; check parser/OCR isolation where available.
- Check HTTPS/TLS 1.2+ enforcement and ensure insecure transport is not accepted.
- Inspect application logs and client errors for credentials, tokens, unnecessary content, or sensitive embeddings.
- Verify audit coverage for access-relevant and administrative actions.
- Check rate limiting and safe errors for malformed or high-volume requests.

6. Test Environment

Area	Planned setup
Hardware	Developer/test workstation (about 4 CPU cores, 8–16 GB RAM, no GPU required) and agreed hosting environment; no dedicated hardware is required.
Software	Application/API build, browser, metadata database, file/object storage, queue/worker, search index, and parser/OCR/model adapters or deterministic stubs.
Tools	Postman or equivalent for API checks; pytest/JUnit or equivalent for unit/integration tests; JMeter/k6 for load testing; browser accessibility tools; issue tracker. Tool choices are provisional.
Accounts	At minimum: general user with access, general user without access to a protected document, content editor, and administrator. Use synthetic identities.
Documents	Synthetic documents in each supported format, scanned samples if OCR is in scope, metadata-rich files, duplicate pairs, malformed files, and varied topics/categories. Proposed size: about 300 labelled documents (PDF/DOCX/TXT) in 5–6 categories, split about 70/30 into training and test sets.
Search corpus	About 20–50 reviewed natural-language queries with relevance labels and relevant/irrelevant documents. Record corpus, model, index, and configuration versions.

7. Test Schedule

The schedule below is relative to build readiness and plan approval; the windows are indicative planning estimates, not claims of completed work.

Milestone	Indicative timing	Output / gate
Review requirements and design test cases	Week 1 after approval	Approved cases, RTM mapping, and benchmark plan.
Prepare test data and environment	Weeks 1–2	Environment checklist, synthetic corpus, service stubs.
Unit and integration testing	Weeks 2–3	Execution evidence and defect records.
System, security, and accessibility testing	Weeks 3–4	End-to-end, security, and accessibility results.
Performance and AI/search quality evaluation	Week 4	Benchmark results with dataset/configuration detail.
Regression and acceptance review	Week 5	Updated RTM, open-defect disposition, test summary and sign-off.

8. Test Deliverables

Deliverable	Stage 1 disposition
Software Test Plan	Included in this submission.
Manual test cases / test case register	Included in Section 13 with expected results and requirement mappings.
Automated test scripts	Future deliverable after implementation language and APIs are finalized.
Test data catalogue	Specified in Section 6; synthetic files to be prepared for execution.
Test execution logs and evidence	Future deliverable; no test has been run in Stage 1.
Defect reports and retest records	Future deliverable if execution identifies defects.
Final Test Summary Report	Future deliverable after planned test execution.

9. Roles and Responsibilities

Role	Name	Responsibility
QA / Test Lead	Shreeverdhan M	Own test plan, case coverage, execution coordination, status, summary reporting, and coordinate specialist security review where available.
Test Engineer	Sripada Krishna Sameer Aditya	Prepare test data, execute cases, capture evidence, log defects, and retest fixes.
Developer	Srivamshi Chikkireddi	Provide testable builds, support diagnosis, fix defects, and supply release notes.
Project / Product Representative	Shamanth SK	Clarify expected behavior, approve acceptance criteria and defect dispositions.

10. Risks and Mitigation

Risk	Mitigation
No representative document corpus	Create diverse synthetic samples and label a query-document relevance benchmark before judging search quality.
AI output varies by model or run	Record model/configuration version, define evaluation data, and treat enrichment as a suggestion.
Delayed build/environment	Prepare the case register and data in advance; do not mark unexecuted cases as passed.
External model/index outage	Use controlled stubs for deterministic tests and separately test timeouts, retries, and failure states.
Metadata/index drift after deletion	Test deletion and re-indexing, compare state across stores, and include reconciliation checks.
Restricted data leaks via search	Prioritize cross-role tests for IDs, titles, snippets, metadata, vectors, and file retrieval.

11. Assumptions and Dependencies

- Supported formats, upload limits, and target languages will be agreed before test execution.
- A stable test build, isolated test environment, safe credentials, and synthetic data will be provided at Stage 2.
- AI/model and search-provider choices may change; acceptance tests should emphasize observable contracts.
- Search quality targets and the benchmark corpus must be agreed before results can be judged.
- The proposed calendar is dependent on build readiness, team availability, and upstream service access.

12. Suspension and Resumption Criteria

12.1 Suspend Testing If

- The test environment is unavailable for more than four working hours and no approved backup is available.
- The build is unstable enough to block more than 30% of planned cases because of a common failure.
- Test data or credentials may expose actual sensitive information.
- A critical defect makes continued testing unsafe or invalidates other results.

12.2 Resume Testing When

- Blocking defects are fixed or an explicitly approved workaround is documented.
- Environment health and smoke checks pass; required dependencies are stable.
- Credentials and test data are verified safe and valid.
- Tests affected by the failure are re-run and previous results are corrected where necessary.

13. Test Case Management and Traceability

Case IDs are stable and match the SRS acceptance references and RTM. Initial status for every case is "Not Run – planned." During Stage 2, record execution date, build/model/index version, tester, actual result, evidence location, and defect ID. Status values: N = Not Run, P = Passed, F = Failed, B = Blocked, W = Waived.

Test ID	Scenario / steps summary	Linked requirement(s)	Expected result	Initial status
TC-UP-01	Upload a valid supported document as an authorized user.	AICMS-F-001	202/accepted document ID returned; status is QUEUED or PROCESSING.	N – planned
TC-UP-02	Upload an unsupported file and a file larger than configured limit.	AICMS-F-002; AICMS-SR-002	Each invalid upload is rejected safely and not indexed.	N – planned
TC-UP-03	Upload the same content twice under configured duplicate policy.	AICMS-F-003	Configured duplicate warning or action occurs; no unexplained duplicate index entry.	N – planned
TC-UP-04	Observe status from upload through completion; simulate processing error.	AICMS-F-004; AICMS-NF-008	States are visible; failure is not marked complete and has a safe retry/error state.	N – planned
TC-EXT-01	Upload text-based supported file with known body text.	AICMS-F-005	Extracted text matches the expected content sufficiently for indexing.	N – planned
TC-EXT-02	Upload document with title/author/date metadata and one without metadata.	AICMS-F-006	Available metadata is stored; missing values remain empty rather than fabricated.	N – planned
TC-EXT-03	Upload supported scan with OCR enabled and a low-quality scan.	AICMS-F-007	OCR text is stored; poor extraction is marked for review/quality warning.	N – planned
TC-AI-01	Process a labelled document whose topic/category is known.	AICMS-F-008	A category suggestion and the chosen model/version or review metadata are recorded.	N – planned
TC-AI-02	Process documents with distinct topics and inspect keywords/topics.	AICMS-F-009	Representative terms/topics are stored; quality is judged against agreed examples.	N – planned
TC-AI-03	Process eligible text and simulate embedding provider failure.	AICMS-F-010; AICMS-NF-008	Embedding/index job succeeds or shows a visible retryable failure state.	N – planned
TC-AI-04	Run the trained classifier on the held-out labelled test set (about 30% of the corpus) and compare suggestions with the reviewed labels.	AICMS-F-008; AICMS-NF-009	At least 80% of test documents receive the correct category; per-category results are recorded.	N – planned
TC-IDX-01	Search for exact and semantic concepts in a successfully processed document.	AICMS-F-011	Document becomes discoverable after indexing completes.	N – planned
TC-SR-01	Submit a valid natural-language query with expected results.	AICMS-F-012	Authorized results or explicit no-results state is returned.	N – planned
TC-SR-02	Use a reviewed query-document relevance set to inspect result order.	AICMS-F-013	Relevant seeded documents rank above unrelated items according to agreed metric.	N – planned
TC-SR-03	Combine category and file-type filters.	AICMS-F-014	Every returned result satisfies all selected filters.	N – planned
TC-SR-04	Inspect result title, snippet, category, missing metadata, and permission rules.	AICMS-F-015; AICMS-SR-004	Fields are clear, safe, and only from authorized documents.	N – planned

Test ID	Scenario / steps summary	Linked requirement(s)	Expected result	Initial status
TC-MD-01	Editor modifies categories/keywords; non-editor attempts the same.	AICMS-F-016; AICMS-SR-006	Authorized update persists and is audited; unauthorized edit is denied.	N – planned
TC-SEC-01	Attempt read/edit/delete via direct document ID without permission.	AICMS-F-017; AICMS-SR-001	Unauthorized operations are denied without revealing protected data.	N – planned
TC-SEC-02	Perform metadata edit, taxonomy change, role change, and deletion.	AICMS-F-018; AICMS-SR-006	Each action produces audit event with actor/time/action/outcome, no secrets.	N – planned
TC-SEC-03	Inspect browser/API and service transport configuration.	AICMS-NF-003; AICMS-SR-003	HTTPS/TLS 1.2+ enforced and insecure transport path disabled.	N – planned
TC-SEC-04	Submit malformed/disguised content and supported large file.	AICMS-SR-002	Unsafe files are rejected/isolated; valid in-limit file remains processable.	N – planned
TC-SEC-05	Search as a user without access to a restricted document.	AICMS-F-017; AICMS-SR-004	Restricted ID, title, snippet, metadata, embedding, and content are not exposed.	N – planned
TC-SEC-06	Inspect server logs and client errors after auth/model/parser failures.	AICMS-NF-006; AICMS-SR-005	No password/token or unnecessary raw document contents; correlation IDs support diagnosis.	N – planned
TC-SEC-07	Inspect the user store, logs and API responses after registering users and logging in.	AICMS-SR-007	Passwords exist only as salted hashes; none appear in logs or responses.	N – planned
TC-SEC-08	Leave a session idle for 30 minutes; then make five consecutive failed logins followed by a sixth attempt.	AICMS-SR-008	Idle session requires re-login; the sixth attempt is blocked temporarily and an audit event is recorded.	N – planned
TC-SEC-09	Log in with valid and invalid credentials; log out; reuse the old session token.	AICMS-F-023; AICMS-SR-001	Valid login creates a session; invalid login shows a generic error; logout invalidates the token.	N – planned
TC-ADM-01	Non-admin and admin each attempt a category taxonomy change.	AICMS-F-019; AICMS-SR-001; AICMS-SR-006	Only authorized admin change succeeds and generates audit event.	N – planned
TC-ADM-02	Admin requests re-indexing following taxonomy/model configuration update.	AICMS-F-020	Job ID/status is returned and eligible documents are queued.	N – planned
TC-ADM-03	Admin creates a user, assigns and changes a role, and deactivates the account; non-admin attempts the same.	AICMS-F-024; AICMS-SR-001	Only admin changes succeed, take effect on the next request, and are audited; deactivated user cannot log in.	N – planned
TC-DATA-01	Delete a document then query matching terms.	AICMS-F-021; AICMS-NF-007	After deletion completes, deleted document no longer appears in active search and action is audited.	N – planned
TC-OPS-01	Simulate parser or model service failure.	AICMS-NF-008	Failure state, safe error, and operational event appear; no false success.	N – planned
TC-OPS-02	Inspect metrics for upload errors, job duration, search latency, and service errors.	AICMS-F-022	Authorized admin can view aggregates without document secrets/content.	N – planned
TC-OPS-03	Evaluate uptime measurement over a representative observation window.	AICMS-NF-002	Availability percentage can be calculated with agreed maintenance treatment.	N – planned
TC-PERF-01	Run the approved natural-language search load against the benchmark corpus.	AICMS-NF-001	90th percentile search response time ≤ 3 seconds under named conditions.	N – planned
TC-PERF-02	Run long AI processing while making independent search/upload requests.	AICMS-NF-004	Long job is queued and unrelated requests remain within agreed responsiveness targets.	N – planned
TC-PERF-03	Upload a 10-page text-based document under normal test load and time extraction, enrichment and indexing to completion.	AICMS-NF-010	Median and p90 processing time ≤ 60 seconds in the named environment.	N – planned
TC-UX-01	Complete upload, search, filter, and metadata review with keyboard only.	AICMS-NF-005	Focus is visible; controls are labelled and core flows can be operated.	N – planned

Test ID	Scenario / steps summary	Linked requirement(s)	Expected result	Initial status
TC-DATA-02	Modify metadata, re-index, delete, and run database/index reconciliation.	AICMS-NF-007	No unexplained stale index entry or orphaned active record remains.	N – planned

13.1 Test Data Catalogue

Data group	Representative cases	Purpose / safety constraints
Valid document corpus	At least two text documents per agreed format, varied topics, metadata present/absent.	Verify extraction, metadata, categorization, topic extraction, and indexing. Use synthetic/public-safe content.
Negative upload samples	Wrong extension/content signature, malformed file, over-limit file, duplicate pairs.	Validate intake safety and duplicate policy; never use malicious production data.
OCR set	Clear scan, skewed/low contrast scan, image-only page.	Check OCR route and quality-warning behavior if OCR is in scope.
Access-control records	User A allowed, User B not allowed, editor, administrator; at least one restricted document.	Exercise least privilege and prevent disclosure of restricted search results.
Search benchmark	Natural-language queries with reviewed relevant and irrelevant documents/labels.	Evaluate ranking with Precision@5/relevant coverage; record reviewer and version.
Failure simulations	Model unavailable, parser error, search index timeout, queue retry.	Verify sanitized errors, retry/failure state, audit/metrics, and no false success.

14. Test Metrics and Reporting

- Execution progress = executed cases / planned cases × 100%.
- Pass rate = passed cases / executed cases × 100%; blocked and waived cases are reported separately.
- Requirement coverage = in-scope requirements mapped to test cases / total in-scope requirements × 100%.
- Defects: count by severity, age, component, reopen/retest status, and requirement impact.
- Search quality: Precision@5, relevant-result coverage (target: relevant document in the top 5 for ≥ 80% of queries), and category/keyword reviewer assessment on the agreed corpus.
- Categorization accuracy = documents with the correct top category / total labelled test documents × 100% (target ≥ 80%).
- Performance: p50/p90 search latency, error rate, queue delay, extraction/enrichment duration, and concurrency profile.
- Reports: execution status, defect summary, security findings, performance/relevance benchmark, and final test summary.

15. Approvals

Role	Name	Signature / Date
QA / Test Lead		
Project Representative		
Course Coordinator		